



Triboelectric-electromagnetic metamaterial systems for self-powered multiparameter wireless monitoring

Wentao Li^{a,b,c}, Kanghui Han^d, Guohui Shen^d, Zhiguo He^{a,b,*}, Pengcheng Jiao^{a,b,*}

^a State Key Laboratory of Ocean Sensing & Ocean College of Zhejiang University, Zhoushan, Zhejiang 316021, China

^b Hainan Institute of Zhejiang University, Sanya, Hainan 572025, China

^c Interdisciplinary Student Training Platform for Marine Areas, Zhejiang University, Hangzhou, Zhejiang 310027, China

^d College of Civil Engineering and Architecture, Zhejiang University, Hangzhou, Zhejiang 310058, China

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ABSTRACT

Self-powered, multiparameter real-time monitoring is essential for ensuring the safety of large-scale structures, influenced by wind disasters such as typhoons. However, it remains inadequate in detecting multiparameter dynamic response of complex large-scale structures under extreme working conditions. Herein, this work develops the wind-driven triboelectric-electromagnetic metamaterial systems to monitor the multiparameter response of large-scale truss structures under strong wind. The hybrid metamaterial systems consist of the rotational and translational components that include the triboelectric and electromagnetic generators, respectively, realizing multiparameter self-powered monitoring of the structure and environments. Experiments are performed to investigate the electrical response of the triboelectric-electromagnetic devices under different rotational speeds, frequencies and wind speeds. The reported devices are then integrated into the wireless monitoring systems to realize real-time detection. Eventually, the metamaterial systems are applied to monitoring the multiparameter response of large-scale transmission tower model. The structural dynamic response (i.e., stress) is monitored by the rotational component, and the environments (i.e., temperature and humidity) are determined by the translational component. The reported metamaterial systems provide novel design guidance for real-time, self-powered multiparameter monitoring of large-scale engineering structures.

1. Introduction

Massive transmission towers, as quintessential large-scale truss structures, form the fundamental framework of the global energy network [1]. Ensuring the safe and stable operation of these structures is imperative, particularly under extreme environmental conditions [2]. However, the routine operation and maintenance of such infrastructures remain challengeable due to their considerable size, remote locations, and complex structural configurations, let alone the traditional manual inspection is typically inefficient and costly [3,4]. For example, transmission towers are subjected to complex environmental loads during long-term services, e.g., strong winds, ice cover, drastic temperature changes, etc. [5]. Accurate real-time monitoring of multiparameter such as temperature and vibration is essential to guarantee the safe functioning of towers [6]. Structural health monitoring (SHM) technologies demand precise data acquisition, minimal power consumption, and reliable long-distance wireless transmission [7,8]. Thanks to the rapid

development of advanced sensing and wireless transmission technologies, integrated SHM systems have become a key alternative to achieve efficient monitoring [9,10]. Comparing with the traditional wireless transmission technologies such as Bluetooth and Zigbee, etc., long range radio (LoRa) has attracted research and practice attention due to their long transmission distance (>1 km) and low power consumption (~mW) [11,12]. However, battery power supply of wireless monitoring system severely suffers from short lifespan and high maintenance cost [13–16]. Therefore, it has become an innovative and feasible technical way to utilize environmental energy for self-powering of wireless monitoring.

Wind energy is characterized by its widespread distribution, extensive availability and high-power density [17,18]. Comparing with solar and mechanical energy solutions, wind energy attracts research interests due to its ability to generate electricity around the clock, low land constraints, and economic viability for grid-scale deployment [19,20]. Recent advancements in wind energy harvesting predominantly focus

* Corresponding authors at: State Key Laboratory of Ocean Sensing & Ocean College of Zhejiang University, Zhoushan, Zhejiang 316021, China.

E-mail addresses: hezhihuo@zju.edu.cn (Z. He), pjiao@zju.edu.cn (P. Jiao).

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on electromagnetic generators (EMG) [21], piezoelectric nanogenerators (PENG) [22], triboelectric nanogenerators (TENG) [23], and hybrid generators systems [24–27]. PENG is inadequate in low frequency wind energy conversion efficiency, limited output power and material brittleness, which hinder their ability to effectively capture irregular breeze energy [28]. TENG has attracted widespread attention in recent years due to its advantages of strong design flexibility, simple structure, and low cost [29–35]. The hybridization of TENG and EMG offers a complementary approach, enhancing the effective operational wind speed range and optimizing electrical output characteristics [36]. Therefore, integrating TENG and EMG devices presents a promising strategy for efficient wind energy harvesting across a broad spectrum of wind speeds. For example, the stackable triboelectric-electromagnetic hybrid nanogenerator was reported to construct a self-powered environmental governance system [37]. The bidirectional rotating turbine hybrid triboelectric-electromagnetic wave energy harvester was developed for marine environment monitoring [38]. However, research on self-powered multiparameter monitoring for large-scale structures is still in its infancy.

In this study, we report the self-powered wireless monitoring systems, driven by a wind-driven triboelectric-electromagnetic metamaterial systems (TE-MS), for multiparameter monitoring of large-scale truss structures. The TE-MS is designed of the rotational and translational components, each containing TENG and EMG for functionalization. We first investigate the electrical responses of the rotational and translational components at different rotational speeds and frequencies. The two components are then connected by a linkage mechanism and the electrical performance is tested at different wind speeds and external resistances. Finally, we develop the integrated wireless monitoring systems powered by the TE-MS to monitor multiple parameters (i.e., structural stresses, temperature and humidity) of a transmission tower. The monitoring results show that the proposed metamaterial systems can accurately capture the dynamic response of the transmission tower and the data can be collected and transmitted wirelessly over long distances in real-time. These findings suggest that the TE-MS systems offer an effective solution for real-time, self-powered monitoring of large-scale engineering structures.

2. Results and discussion

2.1. Application scenarios, structural features and working principle of the TE-MS

Fig. 1A illustrates the distribution of global electricity consumptions in 2024, where the top three countries are China over 32 %, US 14.3 % and Russia 3.9 % [39]. This substantial level of electricity usage underscores the critical needs for transmission towers and the implementation of effective SHM systems. Fig. 1B depicts the evolution of global electricity generation over the past two decades, highlighting the growing significance of wind energy. Specifically, the share of wind power has increased from 0.6 % in 2005–8.1 % in 2024, demonstrating its expanding effect in the energy sector. As a consequence, the effective utilization of wind energy, particularly small-scale distributed wind energy, has become a concern of great interest.

The TE-MS comprises the rotational and translational components, as well as a linkage mechanism, as illustrated in Fig. 1C. The wind cup initiates rotation under the influence of wind force, effectively triggering the rotational component. Subsequently, the linkage mechanism converts this rotational motion into linear movement of the translational component. Fig. 1D and Fig. 1E illustrate the extreme positions of the linkage mechanism. The electrical energy produced by the rotational and translational components is utilized to supply power to the wireless monitoring system and the hygrothermograph, respectively, thereby facilitating the simultaneous monitoring of multiparameter related to both the structure and its surrounding environment. Fig. 1F illustrates the structural composition of the rotational component within the TE-

MS system. The rotational component includes the two power generation units of TENG1 and EMG1. The TENG1 device comprises four brushes arranged at 90-degree intervals, a polytetrafluoroethylene (PTFE) film, and four pairs of electrodes fabricated from the copper foil. The brush is affixed to a rotating shaft to guarantee its synchronous rotation with the wind cup. The PTFE film is adhered to conductive copper foil, and the composite structure is subsequently affixed to the upper surface of a stationary base. Each electrode pair is composed of two neighboring copper foil segments. EMG1 is composed of a magnetic layer and a copper coil layer. The copper coil layer is embedded within the bottom surface of the stationary base, while the magnetic layer is attached to the rotating shaft. Consequently, the rotation of the magnetic layer induces the generation of electrical signals. The composition of the rotational component is shown in Note S1 in the Supplementary Materials.

Working mechanism of TENG1 and EMG1 are illustrated in Fig. 1G. In the TENG1 part, electrostatic induction initially generates equal quantities of positive and negative charges on the surfaces of the PTFE and the brushes, resulting in the induction of corresponding positive and negative charges on the electrode pairs. As the wind cup commences rotation, causing the brushes to slide over the PTFE surface, electrons flow between the electrode pairs through an external circuit up to stage iii. Subsequently, as the brush continues to move forward, electron transfer between the electrode pairs occurs in the opposite direction, ultimately restoring the system to its initial state. In the EMG1 part, based on Faraday's law of electromagnetic induction, the rotation and approach of the magnet toward the copper coil result in a progressive increase in the magnetic flux passing through the coil. This variation in magnetic flux induces an electric current within the coil. Conversely, as the magnet continues its rotation and recedes from the coil, the induced current reverses direction accordingly.

Fig. 1H illustrates the structural composition of the translational component within the TE-MS system. Similar to the rotational component, the translational component includes two power generation units, TENG2 and EMG2. The translation component comprises a hollow stationary base and a reciprocating cylinder. The interior surface of the fixed base is lined with copper foil, while its exterior is wound with copper coils. The reciprocating cylinder is prepared by 3D printing with a magnet inside. Its exterior is coated first with copper foil, followed by an outer layer of PTFE film. The copper foil lining the interior of the fixed base, together with the PTFE and copper foil layers on the exterior of the reciprocating cylinder, constitute the TENG2. Concurrently, the copper coil on the exterior of the fixed base and the magnet embedded within the reciprocating cylinder form the electromagnetic generator EMG2. A photograph of the translational component is shown in Note S2 in the Supplementary Materials.

Working mechanism of TENG2 and EMG2 is shown in Fig. 1I. Initially, in the fully overlapped configuration (I), the PTFE surface acquires a negative charge, while the lower copper foil's upper surface becomes positively charged. Due to electrostatic induction, the upper copper foil remains electrically neutral overall, resulting in a balanced system with no current flow. Subsequently, as the upper structure slides leftward (II), the fixed positive charges on the lower copper foil become misaligned with the fixed negative charges on the PTFE surface. This misalignment enhances electrostatic induction, prompting electrons to flow from the upper copper foil to the lower copper foil. During the reverse sliding motion (IV), electrons flow back from the lower copper foil to the upper copper foil, thereby establishing a reciprocating cycle. In EMG2, according to Faraday's law, as the magnet rotates and moves closer to the copper coil, the magnetic flux through the copper coil gradually increases, and the change in flux causes an induced current to flow through the coil. As the magnet continues to rotate and gradually moves away from the coil, an induced current is generated in the opposite direction.

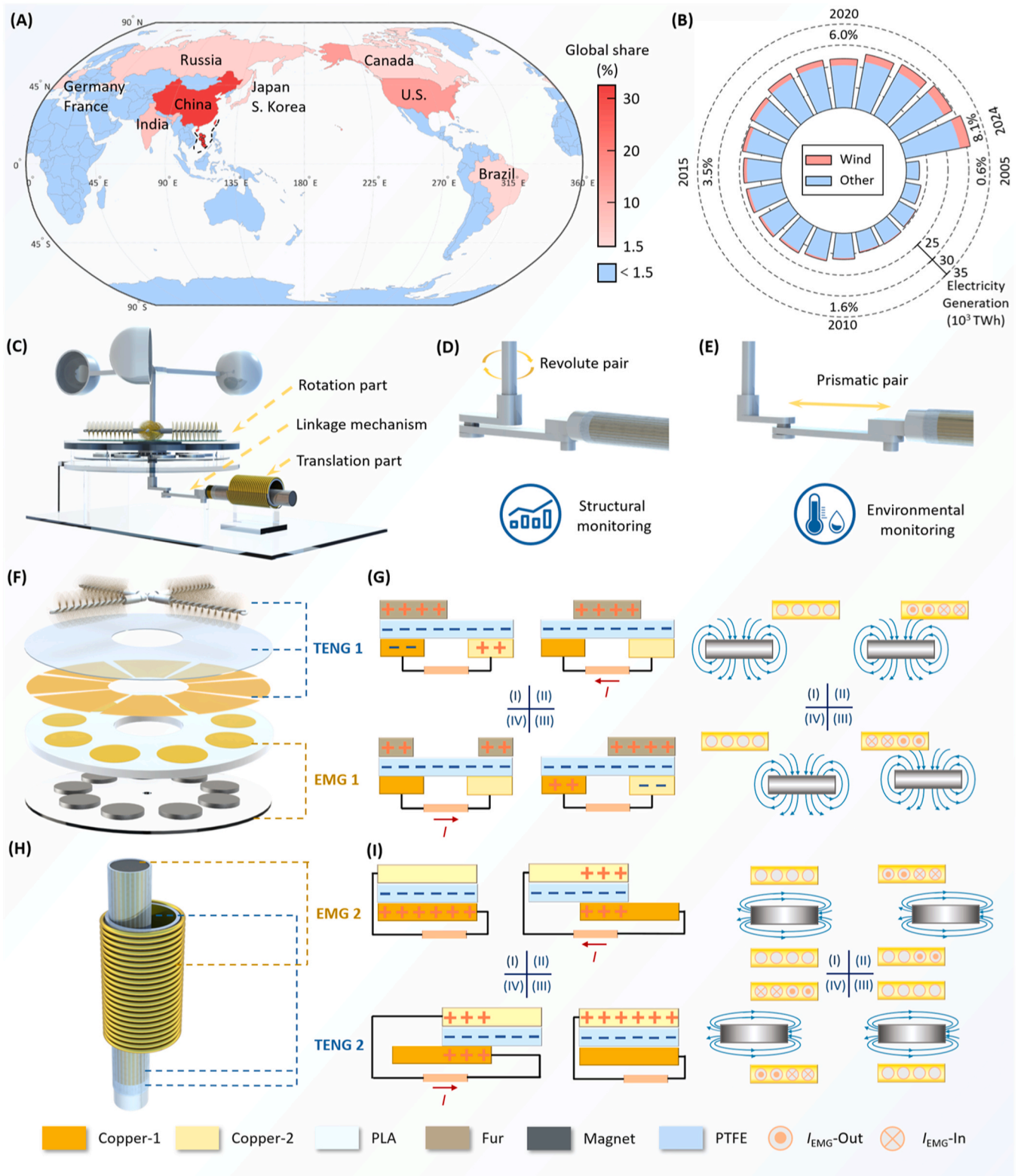


Fig. 1. Application scenarios, structural features and working principle of the TE-MS. (A) Distribution of global electricity consumption by country. (B) Trends in global electricity generation over the past twenty years. (C) Structural design of the TE-MS. (D, E) The extreme positions of the linkage mechanism. (F) Composition of the rotational component. (G) Working principle of the TENG1 and EMG1. (H) Composition of the translational component. (I) Working principle of the TENG2 and EMG2.

2.2. Basic electrical characterization of the components

The electrical properties of the rotational component and the translational component are firstly tested under different conditions using rotary and linear motors, respectively. The schematic diagram of the experimental setup for the rotational component is illustrated in Fig. 2A. The rotary motor is connected to a rotating shaft that rotates the rotational component at different speeds to generate electrical signals. The open-circuit voltage (V_{oc}) and short-circuit current (I_{sc}) of EMG1 at different speeds are shown in Fig. 2B and C, respectively.

When the rotational speed increases, both V_{oc} and I_{sc} of EMG1 tend to increase linearly. Specifically, V_{oc} of EMG1 increases from 1.16 V to 6.67 V as the rotational speed increases from 0.5 r/s to 3 r/s, while I_{sc} increases from 40.3 mA to 241.9 mA. Figs. 2D, 2E and 2F represent the V_{oc} , I_{sc} and short-circuit transferred charge (Q_{sc}) of TENG1 at different rotational speeds, respectively. Different from the results of EMG1, the I_{sc} of TENG1 increase significantly with increasing rotational speed, while V_{oc} and Q_{sc} remains almost constant. Specifically, when the rotational speed increases from 0.5 r/s to 3 r/s, the V_{oc} of TENG1 increases from 218 V to 227 V, the I_{sc} increases from 0.32 μ A to 3.23 μ A, and the

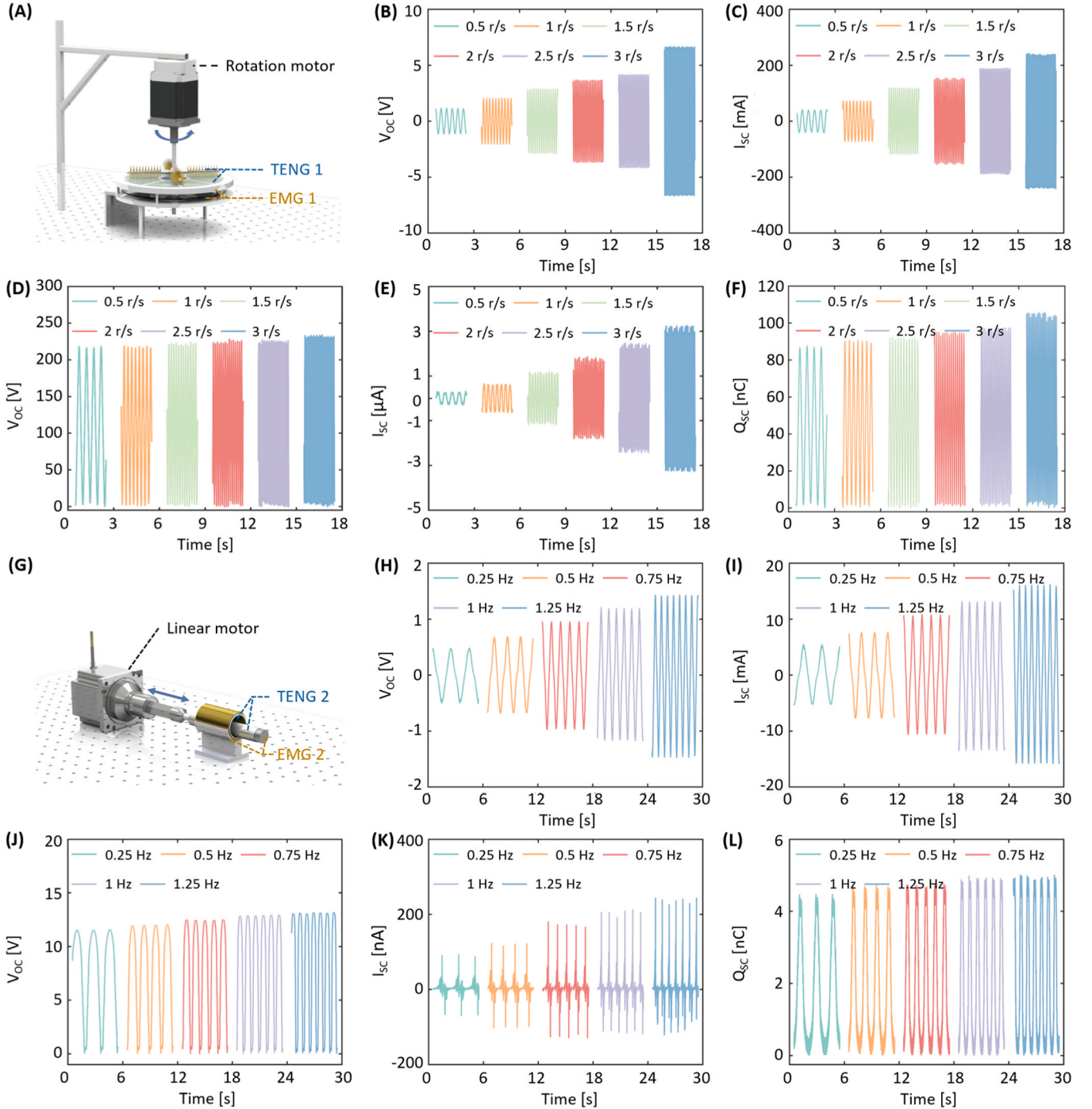


Fig. 2. Basic electrical characterization of the rotational and translational components. (A) Schematic diagram of the experimental setup for the rotational component. Variations of (B) V_{oc} and (C) I_{sc} of EMG1 at different rotational speeds. Variations of (D) V_{oc} , (E) I_{sc} and (F) Q_{sc} of TENG1 at different rotational speeds. (G) Schematic diagram of the experimental setup for the translational component. Variations of (H) V_{oc} and (I) I_{sc} of EMG2 at different reciprocating frequencies. Variations of (J) V_{oc} , (K) I_{sc} and (L) Q_{sc} of TENG2 at different reciprocating frequencies.

Q_{sc} increases from 87.5 nC to 106 nC. This is because the higher the rotational speed, the faster the charge transfer rate and hence the larger I_{sc} . On the contrary, the reason that V_{oc} and Q_{sc} are approximately constant at different rotational speed is that the voltage and transferred charge depends mainly on the effective contact area between the triboelectric pair and is not affected by the rotational speeds.

The schematic diagram of the experimental setup for the translational component is illustrated in Fig. 2G. A linear motor is fixed to the end of the cylinder to drive the cylinder in a regular linear reciprocating motion. The reciprocation range is 50 mm, which is also the reciprocation distance in the TE-MS. The V_{oc} and I_{sc} of EMG2 at different frequencies are shown in Figs. 2H and 2I, respectively. Similarly to the results of EMG1, both V_{oc} and I_{sc} of EMG2 show a good linear increasing relationship as the frequency of reciprocating motion increases. Specifically, V_{oc} of EMG2 increases from 0.48 V to 1.44 V as the frequency

increases from 0.5 Hz to 1.25 Hz, while I_{sc} increases from 5.59 mA to 16.12 mA. The V_{oc} , I_{sc} and Q_{sc} of TENG2 at different frequencies are shown in Figs. 2J, 2K and 2L, respectively. Similar to the results of TENG1, the I_{sc} of TENG2 increase significantly with increasing frequency, while V_{oc} and Q_{sc} remains almost constant. Specifically, when the frequency increases from 0.5 Hz to 1.25 Hz, the V_{oc} of TENG2 increases from 11.6 V to 13.1 V, the I_{sc} increases from 93.6 nA to 244 nA, and the Q_{sc} increases from 4.77 nC to 5.02 nC. For the same reason as TENG1, the rate of charge transfer is positively correlated with frequency, whereas the voltage and transferred charge depend on the contact area rather than the frequency.

Overall, both EMG1 and EMG2 exhibit a linear increase in V_{oc} and I_{sc} with rising rotational speed or motion frequency. In contrast, for TENG1 and TENG2, the I_{sc} increases significantly with speed/frequency, while the V_{oc} and Q_{sc} remain nearly constant. The electrical characteristics of

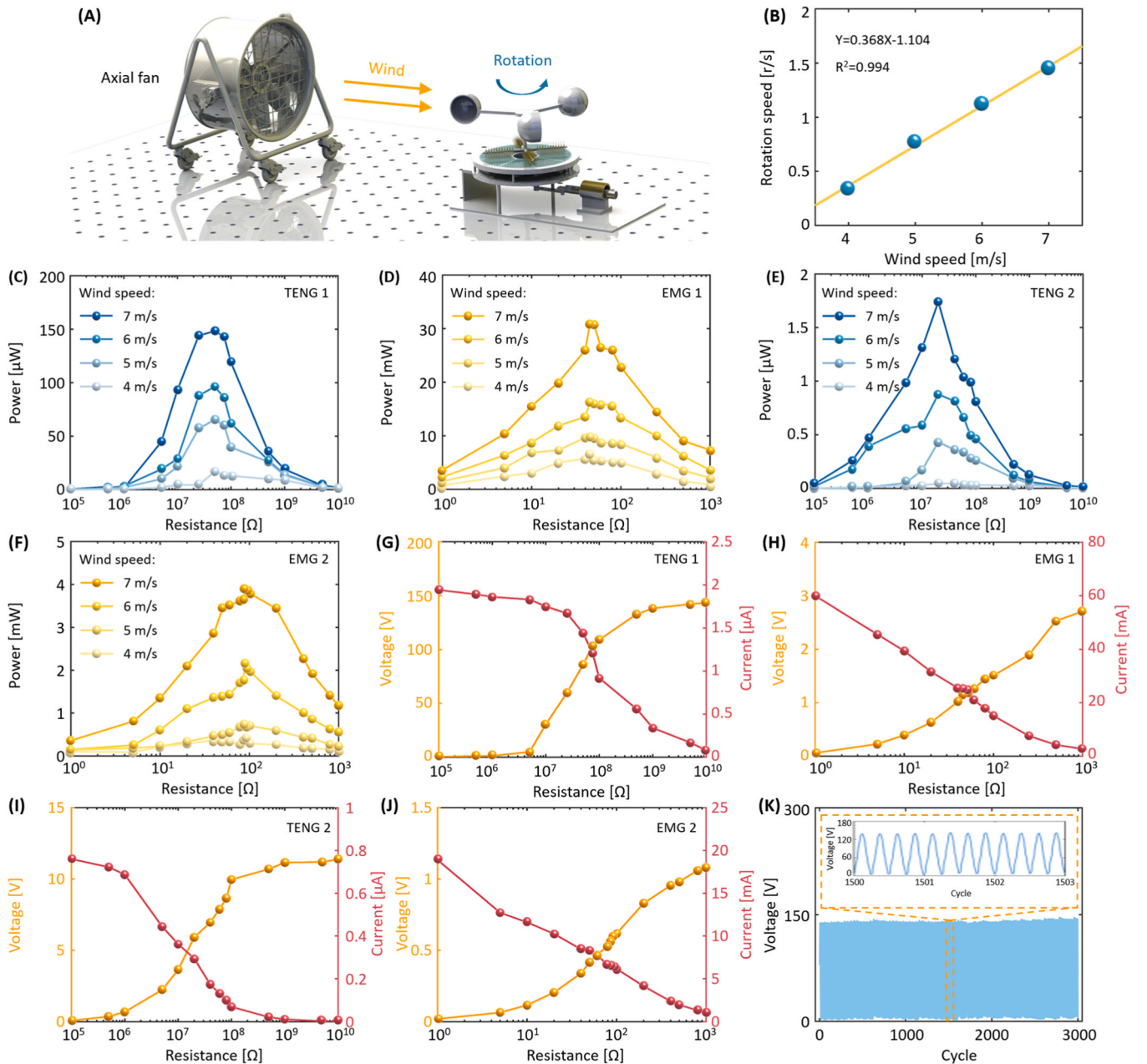


Fig. 3. Electrical performance of the integrated TE-MS device at different wind speeds. (A) Schematic diagram of the experimental setup of TE-MS at different wind speeds. (B) Relationship between the wind speed and rotational speed. Ppeak of TENG1 (C), EMG1 (D), TENG2 (E) and EMG2 (F) for different wind speeds and resistance loads. The variation of voltage and current of TENG1 (G), EMG1 (H), TENG2 (I) and EMG2 (J) under various loads at 7 m/s wind speed. (K) Voltage vs. Cycle for TENG1 and EMG1.

the rotational and translational components in the rotary and linear motors indicate that the TENG and EMG have a more prominent performance at low frequency (i.e., low wind speed) and high frequency (i.e., high wind speed), respectively, which provides the basis for the hybrid energy harvesting.

2.3. Output performance of the hybrid generators

The aforementioned work reveals the performance potential of the rotational and translational component of the TE-MS under idealized excitation conditions using rotary and linear motors, respectively. The

practical application value of the study is ultimately reflected in its output performance driven by real wind inputs, so it is essential to explore the electrical performance of the integrated TE-MS device at different wind speeds. The schematic diagram of the experimental setup for the TE-MS. The schematic diagram of the experimental setup of TE-MS at different wind speeds is shown in Fig. 3A. The TE-MS is placed on a horizontal table where the axial fan can provide a stable wind source to test the output performance of the TE-MS at different wind speeds, and an anemometer is placed at the same horizontal position of the TE-MS to measure the magnitude of the wind speed. The correspondence between the rotational speed and wind speed of TE-MS is firstly investigated and

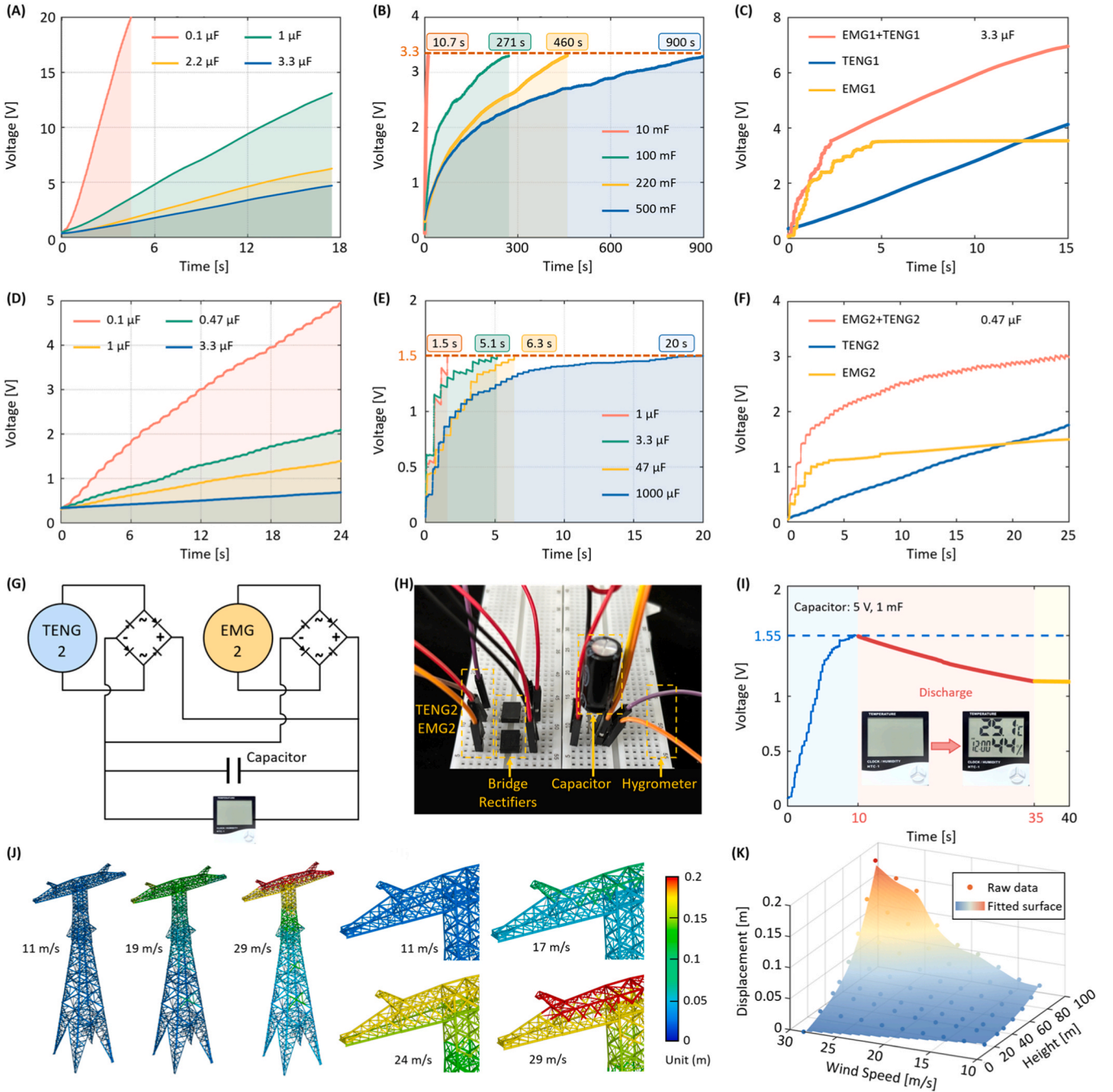


Fig. 4. Energy supply properties and application options of TE-MS. (A) The charging effect of TENG1 (A) and EMG1 (B) for different capacitors. (C) Comparison of the performance of TENG1 and EMG1 with separate power supply and hybrid power supply. The charging effect of TENG2 (D) and EMG2 (E) for different capacitors. (F) Comparison of the performance of TENG2 and EMG2 with separate power supply and hybrid power supply. Schematic (G) and circuit diagram (H) of hybrid energy storage and supply. (I) Effect of hybrid powering of EMG2 and TENG2 for hygrometers. (J) Displacement maps of transmission tower models at different wind speeds. (K) Displacement variations of transmission towers at different heights under different wind speeds.

the results are shown in Fig. 3B. Based on the design of linkage mechanism, the rotation frequency of the rotational component is the same as the reciprocating motion frequency of the translational component, i.e., the rotational component rotates one revolution, the translational component realizes one complete reciprocating motion. At a wind speed of 4 m/s, the TE-MS enables stable driving at a rotational speeds of 0.33 r/s. When the wind speed reaches 7 m/s, the rotational speed of TE-MS becomes 1.45 r/s. It is also found that the rotational speeds of TE-MS show a good linear relationship with the increase of wind speed. The peak power P_{peak} of each generating unit is then tested for different wind speeds and resistance loads R , and the results are shown in Fig. 3C-3F. As the wind speed increases from 4 to 7 m/s, the P_{peak} of TENG1 increases from 17.0528 μ W to 148.26 μ W at a matching R of 50 M Ω (see Fig. 3C). Similarly, the P_{peak} of EMG1 increases from 6.5 mW to 30.94 mW, at a matching R of 45 Ω (see Fig. 3D). The P_{peak} of TENG2 increases from 0.04 μ W to 1.74 μ W, at a matching R of 20 M Ω (see Fig. 3E). The P_{peak} of EMG2 increases from 0.4 mW to 3.89 mW, at a matching R of 90 Ω (see Fig. 3F).

Fig. 3G-3I show the performance of the TE-MS under various loads at a wind speed of 7 m/s. Similarly, as the load increases, the voltages of TENG1, TENG2, EMG1 and EMG2 show the increasing trend while the currents all show a decreasing trend. This is because as the load resistance increases, the total circuit resistance increases. According to Ohm's law, the circuit current decreases. At the same time, due to the weakening of the voltage dividing effect of the internal resistance of the power supply, the voltage divided between the two ends of the load rises accordingly. Subsequently, the TE-MS is tested for durability at 3000 cycles, and the test results of TENG1, for example, are shown in Fig. 3K. The results show that the TE-MS is still able to exhibit satisfactory performance under multiple cycles.

2.4. Energy supply characteristics and application options for the TE-MS

The charging effects of TENG1 and EMG1 are tested for different capacitors, as shown in Figs. 4A and 4B. The results show that TENG1 enables to charge a 3.3 μ F capacitor to 5 V in 18 s. It can be seen from Fig. 4B that EMG1 can charge a 0.5 F capacitor to 3.3 V in 900 s, which is the operating voltage of many microcontrollers. Fig. 4C shows a comparison of the performance of TENG1 and EMG1 powered individually as well as hybrid. The power supply of TENG1 to the capacitor is continuously rising, and TENG1 enables to charge the capacitor at a relatively high voltage all the time due to the high voltage output characteristic of TENG. On the contrary, EMG1 has a high output current, which enables to quickly charge the capacitor to a specific voltage. EMG1 does not perform well with the increasing of the voltage, and therefore, TENG1 and EMG1 are rectified separately and then connected in parallel to charge the capacitor. This approach allows the TE-MS to utilize the advantages of both generators to produce faster capacitor charging.

Similarly, Fig. 4D-4F show the effects of powering different capacitors by TENG2, EMG2 and the hybrid metamaterial systems. The charging curve of TENG2 for capacitors is usually linear due to the higher output voltage characteristic of TENG, as shown Fig. 4D. Fig. 4E shows EMG2 for charging different capacitors to 1.5 V. Since hygrothermograph typically operate at 1.5 V, this demonstrates the ability to monitor the environment parameters. The hybrid energy supply of EMG2 and TENG2 can exhibit better results than the energy supply alone, as shown in Fig. 4F. To demonstrate the monitoring performance of environmental parameters, EMG2 and TENG2 are rectified separately and then connected in parallel to charge the capacitor such that to operate the hygrothermograph. The schematic and circuit picture are shown in Figs. 4G and 4H, respectively. The capacitor selected for the hygrothermograph is 1 mF, and the effect of the hybrid supply of EMG2 and TENG2 is tested at a wind speed of 7 m/s, as shown in Fig. 4I. When the capacitor is charged to 1.5 V, the axial fan is switched off for testing the time the hygrothermograph operates without external excitation. It

takes 10 s to charge 1 mF capacitor to 1.5 V under the wind speed of 7 m/s, which can supply the hygrothermograph to work stably for 25 s.

Fig. 4J investigates the applications of the reported TE-MS in a transmission tower. To unveil the dynamic response of the tower to determine the monitoring location, a tower model is numerically simulated and analyzed under extreme operating conditions using ANSYS. Each member of the tower is modeled via the beam element by BEAM188, and the entire model consists of 1039 nodes and 2594 beam elements. More details are provided in Section 4.4. The displacement diagrams of the transmission tower at different wind speeds are obtained. Fig. 4K displays the raw data and the fitted surface of the displacements. The displacement is increased with wind speed and monitoring location, and the maximum displacement reaches 0.193 m under the wind speed of 28.54 m/s at a height of 10 m.

2.5. Applications

Here, we integrate the TE-MS into the wireless monitoring systems to realize real-time self-powered monitoring. The working flowchart of the integrated wireless monitoring system is shown in Fig. 5 A. The integrated wireless monitoring system consists of three main components: namely the energy management module, the wireless monitoring system and the visualization module. The energy management module is used to power the wireless monitoring system for real-time data acquisition and wireless transmission. The energy management module consists of EMG1, TENG1, rectifier bridge and capacitors for powering the wireless monitoring system. The wireless monitoring system is designed with the functional modules of the data acquisition, microcontroller unit (MCU) and LoRa wireless transmission, where the MCU is the control core, the inter-integrated circuit (I²C) is the communication interface between the MCU and data acquisition module. The universal asynchronous receiver/transmitter (UART) is the communication interface between the MCU and the LoRa wireless transmission module. Perceived electrical signals generated by vibrations of Lead Zirconium Titanate (PZT) sensors adhered to transmission towers are filtered and converted to digital signals by an analog-to-digital converter (ADC). Furthermore, the real-time data visualization software is developed to visualize the data sent from the monitoring terminal to the computer server in a real-time manner. The PC visualization software displays the data sent wirelessly from the monitoring system to the computer server in real time in a visual format.

The test subject for the application is a large-scale T-shaped angle transmission tower, illustrated in Fig. 5B. This tower with a total height of 84.7 m is situated in Deqing County, Huzhou, China, which is part of the Baihetan-Zhejiang Transmission Project. The application tests are carried out on the T-shaped angle tower that is scaled as 1:50 (see Fig. 5C). PZT sensors are installed at the cross-arm location, as shown in Fig. 5Ci. The wireless monitoring system to measure 7 cm by 5 cm is displayed in Figure 5Cii. The energy supply performance of the TE-MS and the functionality of the wireless monitoring system are evaluated at a wind speed of 13 m/s, as shown in Fig. 5D. A 0.5 F capacitor is chosen because it has enough capacity to power the wireless monitoring system for completing data collection and transmission. Experimental results show that the rotational component can charge the 0.5 F capacitor from 0 V to 4.5 V within 480 s (blue line in Fig. 5D). It is important to highlight that the wireless monitoring system operates at a voltage of 3.3 V and requires only 107 s for initial activation. The video of the first charge of the capacitor is shown in Video S1. Subsequently, the wind tunnel is closed, and the wireless monitoring system is activated. Upon activation, the system begins operation, and the capacitor discharges, causing the voltage to decrease from 4.5 V to 2.8 V over a duration of 55 s (red line in Fig. 5D).

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The server successfully receives 1473 data points over a 55 s monitoring period, demonstrating the capability for real-time data acquisition.

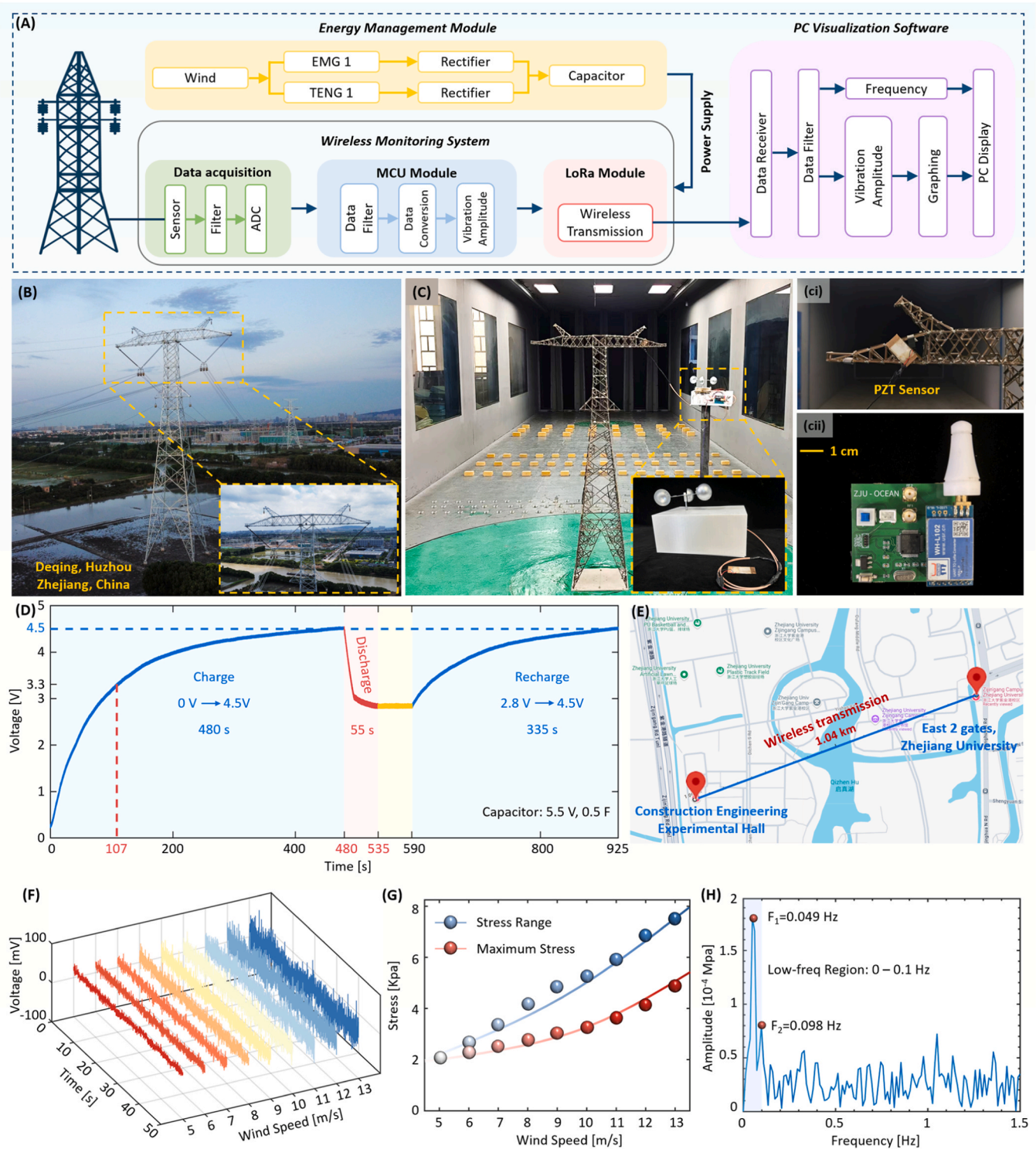


Fig. 5. Application of TE-MS integrated wireless monitoring system. (A) Working flowchart of the integrated wireless monitoring system. (B) Transmission tower used as the test subject and (C) its corresponding model. (D) Charging and discharging curves of capacitors fed by a rotational component. (E) Wireless transmission range of the test data. (F) Sensor raw data recorded at various wind speeds. (G) Comparison of stress ranges and maximum stresses at different wind speeds. (H) Frequency spectrum data at maximum wind speed.

tion and transmission of the wireless monitoring system. A video of the capacitor discharging and the monitoring system starts working is shown in Video S2. Subsequently, the wireless monitoring system is deactivated, and the TE-MS is allowed to remain idle for a designated period (yellow line in Fig. 5D). During this interval, no significant voltage fluctuations are detected. Following this, the wind tunnel is reopened to evaluate the effect of secondary charging on the capacitor.

The findings demonstrate that the capacitor requires only 335 s to charge from 2.8 V to 4.5 V, as illustrated by the blue line on the right side of Fig. 5D. It is also worth noting that charging from 2.8 V to 3.3 V only takes 25 s. Additionally, the wireless transmission distance of the monitoring system is assessed during operation, as depicted in Fig. 5E. The results indicate that the wireless monitoring system supports long-distance transmission exceeding 1 km, thereby offering potential

applications for transmission tower monitoring in field environments. Fig. 5F presents the voltage measurements obtained from the PZT sensor at varying wind speeds. An increase in wind speed corresponds to an amplified impact of wind forces on the structural integrity of the transmission tower. To conduct a more detailed assessment of wind-induced structural damage, the recorded voltage data are transformed into stress values utilizing the piezoelectric effect. Initially, the monitoring data are converted into actual voltage values through a calibration process detailed in Note S3 in the Supplementary Materials. Subsequently, applying the fundamental principles of the piezoelectric effect, the voltage measurements are converted into stress values by $V = Fg_{31}/A$ and $T = F/A$, where V is the true voltage and F is the force applied to the sensor. Since the sensor is adhered to the beam, the force can be the same for both. g_{31} is the piezoelectric constant, t is the piezoelectric sensor thickness, and A is the piezoelectric sensor area. Detailed calculation process is shown in Note S4 in the Supplementary Materials.

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Fig. 5G illustrates the maximum stress and stress range experienced by a transmission tower during a single cycle under different wind speed conditions. Analysis of the monitoring data reveals that as wind speed increases from 5 m/s to 13 m/s, the maximum stress rises from 2.1 kPa to 4.9 kPa, while the stress range expands from 2.1 kPa to 7.8 kPa. These findings demonstrate a pronounced quadratic relationship between wind speed and stress, consistent with theoretical predictions, thereby confirming the reliability of the wireless monitoring system. For further examination, a Fourier transform is conducted on the monitoring data to derive the frequency domain spectrum, as illustrated in Fig. 5H. The frequency domain analysis revealed two prominent peaks at 0.049 Hz and 0.098 Hz, which correspond to the first and second natural frequencies of the transmission tower structure. Moreover, the spectral analysis suggests that the transmission tower model operates within a safe range under wind speeds of 13 m/s. These findings further validate the accuracy of wireless monitoring system.

3. Conclusions

This study developed the wind-driven triboelectric-electromagnetic metamaterial system (TE-MS) for multiparameter monitoring of large-scale truss structures. The TE-MS was designed by the rotational and translational components that were connected by the linkage mechanism. Enabled by the EMG and TENG, the rotational and translational components were used for multi-parameter self-powered monitoring of the structures and environments. The proposed TE-MS was integrated with the wireless monitoring systems to enable real-time acquisition and long-distance wireless transmission of monitoring data. The integrated wireless monitoring system was experimentally validated in a large-scale T-angle transmission tower model. The two components of the TE-MS triggered the hygrothermograph in 10 s for 25 s and the wireless monitoring module in 480 s for 55 s, respectively, realized real-time wireless transmission over the distance of 1.04 km, while the second time to realize the wireless monitoring system working for 55 s was 355 s. The reported integrated TE-MS wireless monitoring system offers a comprehensive framework for real-time, self-powered, and multiparameter monitoring of large-scale truss structures.

4. Materials and methods

4.1. Fabrication of the TE-MS

The housing and linkage of the TE-MS are 3D printed in PLA. The rotational component consists of EMG1 and TENG1. The EMG1 consists of a magnet layer and a copper coil layer. The copper coil layer consists of 10 copper coils sealed externally by resin. The magnet layer consists of 10 permanent magnetic cylinders uniformly distributed in a circular

shape in the plane, with a thickness of 5 mm and a diameter of 20 mm. TENG1 consists of copper foil, PTFE film and a wool brush. The copper foil is divided into eight parts distributed in a circular array, with two neighboring electrodes as opposite electrodes. A 0.05 mm thick PTFE film is fixed to the copper foil to be used as a friction layer, and there are four wool brushes, which are uniformly distributed at 90°, with a diameter of 20 mm and a length of 30 mm. The translational component consists of EMG2 and TENG2. The EMG 2 includes a copper coil and a magnet bar. The magnet bar has a diameter of 15 mm and a length of 60 mm. The TENG 2 consists of copper foil and PTFE film. The materials used for the copper foil and PTFE are the same as those used in TENG1.

4.2. Electrical characterization of the TE-MS

A tensile testing machine manufactured by Han Shen Automation (Jinan) Co., Ltd is used to apply cyclic loading on the translational component for the electrical response in different frequency control mode. The DC speed motors manufactured by Song Yuan Co., Ltd. are used to apply cyclic loads to rotational component in different speed control modes in order to obtain an electrical response. A Keithley System 6514 electrometer is used to record the electrical output, including the open-circuit voltage V_{oc} of the EMG and TENG, and the short-circuit current I_{sc} of TENG, and short-circuit transferred charge Q_{sc} of TENG. The DMM6500 multimeter is used to record the short-circuit current I_{sc} of EMG.

4.3. Wind tunnel testing

The tests are conducted in the ZD-1 Boundary Layer Wind Tunnel at Zhejiang University, with a test section of 4 m in width and 3 m in height. The wind velocity in the wind tunnel can be continuously adjusted from 1 to 55 m/s. The turbulent boundary layers, corresponding to exposure categories B specified in GB 50009–2012, respectively, are simulated with combinations of spires and roughness elements.

4.4. Finite element analysis of the TE-MS

The general finite element analysis software package ANSYS is employed in the study and the finite element model of the tower. Each member of the tower is modeled via the beam element, namely the BEAM188 in the ANSYS element library. The finite element model consists of 1039 nodes and 2594 beam elements. Using harmonic superposition method (HSM) to simulate the time history of wind load pulsation at different positions of the transmission tower. The simulated wind field is exposure categories B specified in GB 50009–2012 code, with a mean wind pressure of 0.45 kN/m² at a height of 10 m. The wind speed spectrum simulated by the HSM method is the Davenport spectrum. The spatial correlation of the fluctuating wind velocities can be characterized by the coherence function and the empirical formula proposed by Davenport is used. The time history of fluctuating wind load is applied to the nodes of the transmission tower. And the transient dynamics analysis method of ANSYS is used to solve the wind-induced response of the finite element model and obtain the displacement response time history.

CRedit authorship contribution statement

Wentao Li: Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis. **Kanghui Han:** Writing – original draft, Visualization, Software, Investigation. **Guohui Shen:** Writing – review & editing, Supervision, Investigation. **Zhiguo He:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Pengcheng Jiao:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.nanoen.2025.111586](https://doi.org/10.1016/j.nanoen.2025.111586).

Data Availability

Data will be made available on request.

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